

EDUCATION

University of Alabama in Huntsville

Aug 2023 – May 2025

Master of Science, Computer Science

- **Thesis:** *Neural Acceleration of Graph Partitioning*

University of Alabama in Huntsville

Aug 2019 – May 2023

Bachelor of Science, Computer Science

SKILLS

- **Programming Languages:** Python, Java, SQL, JavaScript, PHP
- **Frameworks/Tools:** Power BI, PyTorch, Docker, Git/GitHub, Linux, Visual Studio, Pandas, NumPy
- **Databases:** MySQL, MariaDB, SQL Server, SQLite
- **Core Competencies:** Data Analytics, Web Development, REST APIs, Cloud & Containerization

EXPERIENCE

RFPMart LLC

OCT 2025 – Present

Software Developer

- Led early-stage development of a full-cycle ERP procurement platform to streamline compliant procurement workflows.
- Maintained and optimized PHP back-end services and MySQL/MariaDB databases to improve performance and security.
- Developed responsive web applications using USWDS and Bootstrap, ensuring compliance with accessibility standards.
- Delivered technical support and product demonstrations, resolving customer issues and enhancing search performance.

University of Alabama in Huntsville

May 2023 – May 2025

Graduate Research Assistant: ML/HPC

- Developed scalable data pipelines for large graph datasets in HPC environments using Python, enabling efficient preprocessing and feature extraction for ML models.
- Implemented neural network-based approximations for spectral graph partitioning, achieving significant runtime improvements on distributed computing systems.
- Designed and optimized end-to-end machine learning workflows in Python and PyTorch.
- Presented research findings at academic conferences and co-authored a paper submitted to a journal.

KEY PROJECTS

Retail Sales Performance Dashboard | Power BI Desktop, DAX, Power Query

- Built a Retail Sales Dashboard using the Global Superstore Dataset with star schema modeling and KPI reporting.
- Implemented time-intelligence DAX (YoY, rolling revenue), drill-through, and role-level security for actionable insights.

Slide Show Creator | C#, .NET, WPF

- Designed and implemented a desktop application for creating image slideshows with transitions, audio, and local storage.
- Strengthened experience in OO design, GUI development, and maintainable .NET codebases.

Bank Marketing Classifier | Python, SQL, Scikit-Learn, Pandas, Matplotlib

- Developed a predictive model for marketing campaign success using SQL for data extraction and preprocessing.
- Optimized Random Forest with Grid Search & Cross-Validation, achieving 0.95 F1-score.

Multimodal Sentiment Analysis | Hugging Face Transformers, PyTorch, Torchvision, Torchaudio, OpenCV, Pillow, NLTK, SpaCy

- Developed a multi-task deep learning model using PyTorch to classify five lumbar spine degenerative conditions from MRI scans, improving accuracy by 14% through a multi-head architecture and ordinal regression.
- Built an ROI extraction pipeline with OpenCV and SimpleITK based on radiologist annotations, increasing macro F1-score by 11% using masked loss and semi-supervised techniques.

LEADERSHIP/HONORS

- Supercomputing Conference (SC'24) - *Student Volunteer*
- UAH Residence Life - *Resident Assistant*
- UAH Office of Admissions - *Student Orientation Leader*
- Pi Kappa Alpha Fraternity - *Vice President of Membership Development*